

Source-Locked Integration at the First Unsupported Carrier: Cut Archives, Arithmetic Shadows, and a Nonduplicated 1/400 Ledger

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Abstract

TPC-133–140 pursue two branches left open by the MVP3 route audit. The first makes the native entrance executable and follows every atom through a boundary-complete dyadic split to the first carrier not supported by the TPC-17/18 block compiler. The second imports proved fixed-data logarithmic Chowla information and a newly proved small-polylog affine estimate outside a small logarithmic-density exceptional set, isolates a quantitative shift-one non-transfer, and formulates the remaining actual-family and selector estimates needed by the literal determinant-two packet. We integrate these branches without promoting a restricted arithmetic shadow to the physical carrier.

The archive branch gives the exact cut decomposition

$$B_{h_0,\delta}(X) = S_{\text{soft}}(X) + S_{\text{elig}}(X) + S_{\text{front}}(X).$$

The first term is returned as $o(X)$, the eligible tail is sent to the arithmetic corridor, and the frontier remains explicit. Thus a future eligible-tail saving yields only

$$|B_{h_0,\delta}(X)| \leq o(X) + |S_{\text{front}}(X)| + X^{1-\sigma_{\text{elig}}+\Lambda_{\text{elig}}+o(1)}.$$

It cannot close the original packet until the frontier is totalized or separately bounded.

We give a content-addressed manifest, a typed import/export calculus, an exact terminal and tail cover, and an occurrence-level endpoint registry. The MVP1 physical threshold remains the strict amplitude inequality $\Lambda_{\text{phys}} < 1/400$. Unknown costs are not set to zero, determinant reserve is not endpoint slack, and a conservative upper bound at or above the threshold is not a route obstruction without a lower certificate. In the frozen snapshot the first missing record is `H1.frontier_totalization`, the physical registry is incomplete, and no positive L2 estimate is present. This is an L0/L1 integration theorem, not a prime-pair theorem.

Keywords: fixed shift; source-locked manifest; native archive; unsupported carrier; arithmetic shadow; endpoint certificate.

Claim boundary. The machine audit checks hashes, types, exact finite covers, graph syntax, and rational bookkeeping. It does not prove the mathematical correctness of a cited source theorem, the completeness of an arithmetic family, a Möbius or Liouville power saving, or $B_{h_0,\delta}(X) = o(X)$. In particular, “fixed form,” “logarithmic,” and “almost all scales” are not synonyms for the actual growing, all-prefix physical H3 target.

1 The inherited target and the two new branches

Fix a prescribed nonzero even h_0 , a fixed rational $0 < \delta < 1/2$, and a fixed compactly supported smooth weight W . The imported TPC-15 identity writes the weighted fixed-shift defect as

$$\Delta_{h_0,W}(X) = B_{h_0,\delta}(X) + O_A\left(X(\log X)^{-A}\right) \tag{1}$$

for every fixed $A > 0$ [14]. The task is therefore to bound the literal $B_{h_0, \delta}$, not a support-only or averaged-shift surrogate.

TPC-MVP3 reported the requirement for a complete actual-carrier native archive as the first missing object [5]. TPC-133–136 now move that pointer forward, but do not yet satisfy the complete-carrier requirement. TPC-133 gives a canonical tuple generator for the opened atoms; TPC-134 gives an exact dyadic prefix–tail path archive; TPC-135 partitions the archive into the domain eligible for the TPC-17/18 compiler and a geometric frontier; and TPC-136 reconnects the resulting three cut-terminal classes exactly [6–9].

In parallel, TPC-137–140 test whether proved two-point Chowla information can feed the determinant-two arithmetic packet [10–13]. The formal fixed-affine logarithmic baseline is supplied by Tao’s two-point theorem [3]. Quantitative shift-one results of Helfgott–Radziwiłł and Pila are stronger in their stated scope [1, 2], but a shift-one bound is not imported as an estimate for the active odd determinant-two progression. TPC-138/139 additionally record the Tao–Teräväinen small-polylog affine theorem at almost all scales. That theorem is a genuine positive arithmetic shadow, but its exceptional-scale and reduced-component scope is retained exactly.

2 Source-locked records

Definition 2.1 (Literal scope). A scope record is

$$\mathfrak{s} = (h_0, \delta, W_{\text{env}}, \mathfrak{q}, \Omega, \nu, \mathfrak{a}),$$

where W_{env} is the effective support envelope, $\mathfrak{q}$ records all growth, scale, prefix, modulus and weight quantifiers, Ω is the carrier, ν is the single normalization, and $\mathfrak{a}$ is the archive identity. A minimal nonzero support is not inferred from an effective envelope.

Definition 2.2 (Export and import). An export record contains a statement hash, source bundle hash, status, evidence level, direction, coverage, scope, carrier and normalization. A proof import is callable only when those object fields agree exactly or a separately sourced crosswalk theorem is listed. A shadow-only import may deliberately have a different scope, but it has no proof edge to the actual target.

Tuple hashes are integrity checks only. The native tuple itself is the key, and injectivity is checked on the tuples rather than inferred from collision resistance.

Theorem 2.3 (Source-locked non-escalating merge). *Let a finite collection of paper exports and imports pass the following checks:*

- (i) *every proof import has one exact export or a theorem-backed crosswalk;*
- (ii) *the logical dependency graph is acyclic and every primitive node has an independent source;*
- (iii) *no primitive source assumes the hard-packet target or an equivalent target gate;*
- (iv) *terminal leaves, tail occurrences and physical occurrences have exact nonoverlapping covers; and*
- (v) *the declared status and evidence level of every merged node are no stronger than those of its source.*

Then the merged root implication is a valid consequence of the source theorems in their recorded scopes. The merge cannot turn a conditional interface, logarithmic shadow, finite regression or scope-mismatched theorem into a proved growing physical estimate.

Proof. Exact import matching makes every logical edge a typed invocation. Induction along the topological order proves each nonprimitive node from its already justified parents. The primitive-source firewall excludes target-as-premise circularity. Exact covers prevent omitted or duplicated terms, while source-level monotonicity prevents an administrative merge from strengthening a theorem. A shadow-only edge is excluded from the logical graph and hence cannot justify the actual target. $\square$

3 The complete cut and the frontier

Let $\mathcal{A}_X$ be the support-envelope atom archive generated at the native entrance. The TPC-134 dyadic partition and deterministic prefix cut assign every atom to exactly one terminal type:

$$\text{eligible-prefix}, \quad \text{eligible-tail}, \quad \text{frontier}.$$

Column conservation through the cut gives the exact identity

$$B_{h_0, \delta}(X) = S_{\text{soft}}(X) + S_{\text{elig}}(X) + S_{\text{front}}(X). \quad (2)$$

The imported TPC-17 theorem, attached by TPC-136 to the complete eligible-prefix outer sum, gives

$$S_{\text{soft}}(X) = o(X). \quad (3)$$

Proposition 3.1 (Eligible-only compiler obstruction). *The TPC-135 low- L witness is a formal support-envelope native column in the frontier for all sufficiently large X , and its dyadic path-matrix column has sum one. Its full arithmetic coefficient may nevertheless vanish. Consequently the eligible-only coefficientwise compiler is not a complete cover of $\mathcal{A}_X$.*

This is a scoped L1 negative result. It refutes neither the scalar estimate $B_{h_0, \delta} = o(X)$ nor a frontier-augmented signed route. In particular, S_{front} cannot be called soft merely because the eligible-prefix sum satisfies (3).

Definition 3.2 (Frontier totalization). The node `H1.frontier_totalization` is satisfied if every frontier terminal has complete Q_D, Q_Z, G , fixed- h_0 , cover, reconnection and physical-occurrence records, or if one separate theorem bounds the complete original-scale frontier sum by $o(X)$.

4 What the arithmetic shadow proves

TPC-137 combines the formal fixed-affine logarithmic theorem with a fixed periodic/squarefree closure. Its quantifier order freezes the affine forms, periodic modulus and auxiliary cutoff before the asymptotic limit. This is an unconditional arithmetic theorem in a restricted scope, but it is not the growing physical H3 family.

TPC-138 records quantitative shift-one information and proves a determinant/parity non-transfer for the active odd determinant-two carrier. Thus no direct edge

$$\text{quantitative shift one} \longrightarrow \text{actual determinant-two H3}$$

appears in the proof DAG. TPC-139 then names the missing uniform envelope in coefficient height, CRT cutoff, modulus and period. It also records a proved power-of-log estimate for positive nonparallel affine coefficients and constants of sufficiently small polylogarithmic height, outside a small logarithmic-density exceptional set. This restricted result has no automatic edge to the actual carrier: containment of every reduced CRT component, complete squarefree/periodic reassembly, and exceptional-set control remain unproved. TPC-140 separately proves that an almost-scale theorem does not control a prescribed sequence of actual prefixes, formulates the remaining pointwise-or-selector gate, and gives a conditional raw-saving ledger. It also records the exact global-to-window firewall

$$m_{x, \omega}(\mathcal{E}) \ll \min \left\{ 1, \frac{(\log x)^{1-c_0}}{\log \omega} \right\}.$$

Thus a cumulative exceptional-set estimate cannot simply be copied as the same saving on every actual terminal window.

Proposition 4.1 (No shadow teleportation). *A fixed-data logarithmic estimate, a quantitative shift-one estimate, or a small-polylog affine estimate outside a small logarithmic-density exceptional set does not supply the actual H3 packet estimate without, respectively, a growing-family uniformity theorem, a determinant-two transfer, and proved actual-component containment plus squarefree/periodic reassembly and an independent local exceptional-window estimate together with a selector or all-prefix theorem.*

The power-ledger value of both the qualitative logarithmic shadow and the new power-of-log almost-scale shadow is zero on the fixed X -power scale. This means only that neither supplies a fixed power of X ; it is not a lower bound showing that the actual family cannot have a positive power saving.

5 Cut-aware synthesis

Theorem 5.1 (Cut-aware eligible-corridor bound). *Assume that the eligible nonsoft packets admit an amplitude estimate*

$$\max_{b \in \mathcal{B}_X^{\text{elig}}} |\mathcal{P}_b| \leq X^{1-\sigma_{\text{elig}}+o(1)}$$

and that their theorem-backed physical synthesis satisfies

$$\left| \mathcal{S}_X^{\text{elig}}(\{\mathcal{P}_b\}) \right| \leq X^{\Lambda_{\text{elig}}+o(1)} \max_b |\mathcal{P}_b|.$$

Then

$$\boxed{|B_{h_0,\delta}(X)| \leq o(X) + |S_{\text{front}}(X)| + X^{1-\sigma_{\text{elig}}+\Lambda_{\text{elig}}+o(1)}}.$$

In particular, $\sigma_{\text{elig}} > \Lambda_{\text{elig}}$ closes the eligible corridor but does not imply $B_{h_0,\delta} = o(X)$ unless $S_{\text{front}} = o(X)$ or the frontier is included in a complete joint packet theorem.

Proof. Insert (3) into (2); apply the triangle inequality and then the two assumed eligible-corridor bounds. The frontier is an independent summand and remains in the conclusion. $\square$

Remark 5.2 (Relation to TPC-131). TPC-131 proves the complete-carrier formula

$$|B_{h_0,\delta}(X)| \leq o(X) + X^{1-\sigma_{\text{raw}}+\Lambda_{\text{phys}}+o(1)}$$

under a complete archive and registry [4]. [Theorem 5.1](#) is its deliberately incomplete cut-level specialization: it displays the term that prevents invoking that complete theorem.

6 The nonduplicated 1/400 ledger

The fixed MVP1 physical contract is

$$\sigma_{\text{target}} = \frac{1}{400} \quad \text{on the amplitude scale,} \quad \Lambda_{\text{phys}} < \frac{1}{400}.$$

Every physical occurrence is identified by path and stage. An energy exponent e enters the amplitude ledger as $e/2$. A joint theorem replaces a disjoint exact cover of primitive occurrences; it is not retained together with its dependencies.

Four residual namespaces remain distinct:

$$\begin{array}{ll} \text{H3.squarefree_tail,} & \text{H5.content_remainder,} \\ \text{Phys.high_ultra_boundary,} & \text{Archive.frontier_totalization.} \end{array}$$

The selector cost, if any, must be assigned once to either the raw arithmetic crosswalk or the physical synthesis map according to its actual domain and codomain.

Proposition 6.1 (Endpoint certificate directions). *For a fully specified route cell:*

- (a) *a theorem-backed complete upper certificate $U < 1/400$ is a strict physical pass;*
- (b) *a theorem-backed lower or exact dual certificate $L \geq 1/400$ stops that route cell, with equality included; and*
- (c) *an unknown token, an incomplete cell cover, or merely an upper bound $U \geq 1/400$ without a lower obstruction is not a stop certificate.*

The determinant-zero reserve is not a negative physical token.

The current frontier, cover, localization, final reconnection and outer packet-census occurrences are unknown. Consequently

$$\Lambda_{\text{phys}}^{\text{upper}} = \Lambda_{\text{phys}}^{\text{lower}} = \text{not certified,}$$

and the endpoint state is INCOMPLETE. Assigning zero to those occurrences would change the route.

7 The source-locked manifest

The generated manifest contains:

- content hashes for every source bundle TPC-133–140;
- four explicit scope records, including fixed physical, frozen logarithmic, quantitative shift-one and actual H3 target scopes;
- typed exports, proof imports and shadow-only imports;
- an exact proof DAG and declared topological order;
- exactly three terminal roles and four tail occurrences;
- an occurrence registry with acyclic exact-cover replacements;
- the rational endpoint contract and unknown-token state;
- selected-route and scoped stop records; and
- a projection to H1–H9.

The manifest identity is its canonical content, not a self-referential repository commit. Source hashes detect drift but do not prove a theorem or the uniqueness of an atom key.

Table 1: Integrated status after TPC-133–140.

Record	Status	Meaning
Native entrance through cut	PROVED, L1	Exact support-envelope enumeration, path conservation and cut reconnection.
Eligible-only compiler	scoped negative	Refuted as a coefficientwise full cover by a formal support-envelope frontier column; its full arithmetic coefficient remains undecided.
Frontier totalization	NOT-TESTABLE	Downstream maps or a complete original-scale frontier theorem are absent.
Fixed-data logarithmic closure	PROVED, shadow	Arithmetic theorem in frozen logarithmic scope; no growing physical promotion.
Small-polylog affine almost-scale corridor	PROVED, shadow	Positive power-of-log cancellation on eligible reduced components; actual containment, reassembly and deterministic prefixes remain open.
Growing affine uniformity	OPEN, L2 target	Actual coefficient height, masks, weights, phases and prefixes remain uniformly uncontrolled.
Selector/all-prefix return	OPEN	The pulled-back exceptional union is not controlled on every terminal window, and almost-scale information has not been returned to every actual prefix.
Physical endpoint	incomplete	Required occurrence tokens are unknown; no strict 1/400 certificate.

8 Deterministic audit and claim boundary

The standard-library script `experiments/tpc141_batch_integration_audit.py` writes the frozen manifest and an audit report by default. Its `-check` mode performs no writes and compares both artifacts byte for byte. Mutation regressions reject:

1. a proof import with a mismatched shift, carrier or normalization;
2. a duplicate or omitted terminal role;
3. assigning the frontier to the proved-soft destination;

4. merging two tail namespaces;
5. duplicate, self-dependent or overlapping joint tokens;
6. filling an unknown endpoint token with zero;
7. subtracting determinant reserve from the physical loss;
8. a positive-L2 label on a fixed/logarithmic/almost-scale export; and
9. a direct or indirect H9 physical-registry dependency on an arithmetic node.

Established. The source-locked merge theorem, exact cut-aware synthesis, certificate language, and endpoint decision directions are L0/L1 results. The integrated source records also include the restricted small-polylog arithmetic shadow and the scoped eligible-only and shift-one non-transfer obstructions.

Not established. The paper proves no frontier bound, complete actual-carrier H1, growing H3/H5 input, physical H6–H9 conjunction, strict endpoint, $B_{h_0,\delta} = o(X)$, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem. Its first missing node is

H1.frontier_totalization.

9 Conclusion

TPC-133–140 advance the route in two honest ways. The native archive is no longer generically absent: it reaches an exact, reconnected cut and identifies the first unsupported carrier. Separately, known arithmetic cancellation now occupies a precisely typed fixed or small-polylog almost-scale shadow scope, with the actual-family containment, reassembly and selector gaps exposed. Neither result is promoted to the complete physical coefficient.

The remaining frontier term and unknown occurrence tokens force an incomplete endpoint. The correct next action is to totalize or bound the frontier before describing the program as being at a purely arithmetic frontier. If that succeeds, the same manifest can be refrozen and passed to the MVP4 decision calculus without changing the evidence rules.

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